

Michelle D. Davies

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Senior Data Engineer with experience designing and leading large-scale data pipeline architectures, integrating ETL and ML workflows, software development, solution architecture, developer advocacy, technical writing, and mentoring engineering teams. Skilled in Appian, IBM DataStage, Python, TensorFlow, and cloud-based analytics (Azure, AWS). Passionate about optimizing data infrastructure to accelerate machine learning model delivery and insight generation.

Education

University of California, Berkeley – Master of Information & Data Science, 2025

Cornell University, College of Engineering – Bachelor of Science in Electrical and Computer Engineering, 2022

Experience

IBM: Advisory Data Engineer; Cambridge, MA – January 2026 - Present

- Orchestrated end-to-end ETL frameworks using IBM DataStage to manage complex structured and unstructured data flows for AI environments.
- Integrated Appian BPM applications with enterprise data systems to enhance operational efficiency and AI integration.
- Championed technical excellence by mentoring junior staff and enforcing rigorous standards for version control and data architecture.
- Architected ML-ready data pipelines, leveraging Python, R, and TensorFlow to streamline feature engineering and model deployment.

IBM: Senior Data Engineer; Cambridge, MA – April 2024 - December 2025

- Spearheaded full-lifecycle ETL architecture using IBM DataStage, automating the delivery of massive datasets to ML and analytics platforms.
- Integrated Appian BPM solutions with enterprise systems to accelerate workflows and empower AI-backed decision-making.
- Drove operational efficiency by automating workflows, which slashed latency and reduced costs through proactive performance tuning.
- Ensured 24/7 system reliability through dedicated production support, resolving critical incidents to maintain 99.9% data availability.

IBM: Data Engineer & BPM Application Developer; Cambridge, MA – October 2023 - April 2024

- Designed, developed, and deployed business process management applications using Appian (Full-Stack Software Development).
- Designed, developed, and deployed ETL pipeline applications using IBM DataStage.
- Created intuitive and user-friendly interfaces for BPM applications to facilitate easy navigation and use by non-technical users.
- Developed and optimized data processing pipelines using big data technologies to cleanse, structure, and enrich raw data into usable formats.
- Integrated new technologies and data management tools into the existing infrastructure to improve data processing and analytics capabilities.

IBM: Associate Data Scientist; Cambridge, MA – October 2022 - October 2023

- Wrote programs to cleanse and integrate data in an efficient and reusable manner using leading-edge and open-source tools such as Python, R, and TensorFlow, combined with IBM tools and our AI application suites.

- Worked in an Agile, collaborative environment, partnering with other scientists, engineers, consultants, and database administrators of all backgrounds and disciplines to bring analytical rigor and statistical methods to the challenges of predicting behaviors.
- Communicated with internal and external clients to understand and define business needs and appropriate modeling techniques to provide analytical solutions.
- Assisted in leading workshops to gather insight from users about how to improve applications.
- Evaluated modeling results and communicated the results to public sector audiences.

IBM: Client-Facing Technical Support Developer; Littleton & Lowell, MA – May 2021 - December 2021

- Resolved support requests and debugged issues with the IBM InfoSphere Information Server (IIS) product.

Academic & Technical Projects

RealPage Rent Impact, University of California-Berkeley Capstone

Skills: SQL · Python · Pandas · Data Engineering · Business Intelligence

Analyzed 100M+ rental listings with SQL and Python to uncover pricing anomalies linked to RealPage's rent-optimization algorithm.

Context-Enriched NER for Identifying Emerging Trends in Video Comments, University of California-Berkeley

Skills: NLP · Deep Learning · Transformers (BERT) · Python · PyTorch

Fine-tuned BERT models to extract emerging trends and locations from YouTube comments using NLP and PyTorch.

Applications of 5G Cellular Communication Systems, Cornell University

Skills: MATLAB · Communication Engineering

MATLAB simulations compare 5G and LoRaWAN for IoT healthcare, analyzing bit error rate, latency, retransmissions, and reliability, alongside practical and socio-economic impacts.

Machine Learning Systems Engineering: Deploying an API, University of California-Berkeley

Skills: ML Ops · Docker · FastAPI · Model Deployment · Cloud (AWS)

Built and deployed a containerized ML prediction API with FastAPI, Docker, and cloud infrastructure.

Non-Linear Regression for Housing Prices, University of California-Berkeley

Skills: Regression Analysis · scikit-learn · XGBoost · Feature Engineering · Model Evaluation

Compared linear vs. non-linear models (e.g. XGBoost) to boost R^2 on real estate datasets.

Certifications & Digital Credentials

- IBM Data Engineering Specialty - Experienced
- Appian Certified Senior Developer
- Azure AI Fundamentals Certification
- Azure AI Engineer Associate Certification
- AWS Partner Accreditation (Technical)
- AWS Cloud Quest: Cloud Practitioner
- AWS Cloud Quest: Generative AI Practitioner
- Databricks Generative AI Fundamentals Accreditation
- IBM Information Server - Datastage ETL Developer
- IBM Quantum Machine Learning